

B.Tech IV Year I Semester (R20) Regular &amp; Supplementary Examinations December/January 2024/2025

**PRINCIPLES OF COMMUNICATION SYSTEMS**

(Common to CSE, IT, AI&amp;DS, CSE(AI), CSE(DS), CSE(AI&amp;ML), CSE(IoT), CSE(CS), CS&amp;D and AI&amp;ML)

Time: 3 hours

Max. Marks: 70

**PART – A**  
(Compulsory Question)

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- 1 Answer the following: (10 X 02 = 20 Marks)
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|---------------------------------------------------------------------------------|----|
| (a) Define SNR and figure of merit of a communication system.                   | 2M |
| (b) Draw the square law modulator.                                              | 2M |
| (c) With necessary equation, define transmission efficiency of an AM wave.      | 2M |
| (d) Why pre-emphasis and de-emphasis are needed in F.M but not in A.M? Explain. | 2M |
| (e) Define TDM.                                                                 | 2M |
| (f) Why noise immunity of PWM is better than that of PAM?                       | 2M |
| (g) Draw the power spectral density of BFSK signal.                             | 2M |
| (h) What is signal constellation diagram and describe its purpose?              | 2M |
| (i) List the applications of RADAR.                                             | 2M |
| (j) What is frequency reuse?                                                    | 2M |

**PART – B**

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 With block diagram, explain the principle of operation of radio transmitter and receiver. 10M
- OR**
- 3 (a) Define Amplitude modulation and explain about modulation index. 5M  
(b) Explain the principle of V.S.B Transmission. What are its advantages over S.S.B? 5M
- 4 (a) Derive the bandwidth of tone modulated FM signal and the effect of modulation index on bandwidth. 5M  
(b) Draw the phasor diagram of narrow band FM signals with equation. 5M
- OR**
- 5 Formulate the equation for FM wave. Define modulation index, maximum deviation and bandwidth of a FM signal. 10M
- 6 Explain, how a PPM signal can be generated from PWM signal. 10M
- OR**
- 7 Define Pulse Amplitude Modulation (PAM)? Give merits and demerits of PAM. 10M
- 8 Explain in detail about Quadrature phase shift keying. 10M
- OR**
- 9 (a) Discuss the generation and demodulation of binary frequency shift keying. 5M  
(b) Illustrate the principle of regenerative repeater. 5M
- 10 Write about optical communication with block diagram. 10M
- OR**
- 11 (a) Explain the principle of operation of Radar system. 5M  
(b) Write short notes on TDMA and CDMA. 5M

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B.Tech III Year II Semester (R19) Supplementary Examinations November/December 2024

**PRINCIPLES OF COMMUNICATION SYSTEMS**

(Common to CSE, IT, CE, FT, EEE and ME)

Time: 3 hours

Max. Marks: 70

**PART – A**  
(Compulsory Question)

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- 1 Answer the following: (10 X 02 = 20 Marks)
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|----------------------------------------------|----|
| (a) Define SSB and DSB modulation.           | 2M |
| (b) Define Frequency division multiplexing.  | 2M |
| (c) Define angle modulation.                 | 2M |
| (d) Define modulation and demodulation.      | 2M |
| (e) What is meant by Pulse width modulation? | 2M |
| (f) State Sampling theorem.                  | 2M |
| (g) Define amplitude shift keying.           | 2M |
| (h) What is meant by Phase shift keying?     | 2M |
| (i) Write the types of Satellites.           | 2M |
| (j) What is meant by RADAR?                  | 2M |

**PART – B**

(Answer all the questions: 05 X 10 = 50 Marks)

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|-----------|------------------------------------------------------------------------------------------------|----|
| 2         | (a) Explain about DSB-SC modulation.                                                           | 5M |
|           | (b) Explain the need of VSB modulation.                                                        | 5M |
| <b>OR</b> |                                                                                                |    |
| 3         | (a) Explain the generation of AM SSB modulated and demodulation.                               | 5M |
|           | (b) Draw AM Radio Transmitter Block diagram and explain each block.                            | 5M |
| 4         | (a) Discuss about Tone modulated FM signals in detail.                                         | 5M |
|           | (b) Explain about Stereophonic FM Broadcasting in briefly.                                     | 5M |
| <b>OR</b> |                                                                                                |    |
| 5         | (a) Explain about FM Modulation and Demodulation.                                              | 6M |
|           | (b) Comparison between Phase and Frequency modulation.                                         | 4M |
| 6         | (a) Discuss and derive sampling theorem.                                                       | 4M |
|           | (b) Explain about Pulse width modulation.                                                      | 6M |
| <b>OR</b> |                                                                                                |    |
| 7         | (a) Discuss about the Digital representation of analog signals.                                | 5M |
|           | (b) Explain pulse amplitude modulation in detail.                                              | 5M |
| 8         | (a) Explain about Binary Phase shift keying.                                                   | 5M |
|           | (b) Discuss about Binary amplitude shift keying.                                               | 5M |
| <b>OR</b> |                                                                                                |    |
| 9         | (a) Explain about Quadrature phase shift keying.                                               | 5M |
|           | (b) What is Regenerative Repeater explain in detail?                                           | 5M |
| 10        | (a) Draw the block diagram of satellite communication and explain each block.                  | 4M |
|           | (b) Explain the basic principle and operation of Radar. Write the applications of Radar.       | 6M |
| <b>OR</b> |                                                                                                |    |
| 11        | (a) Draw the Block Diagram of Optical Fiber Communication System.                              | 4M |
|           | (b) What are the major elements of earth station? Explain briefly in satellite communications. | 6M |

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B.Tech III Year I Semester (R20) Regular &amp; Supplementary Examinations November/December 2024

**PRINCIPLES OF COMMUNICATION SYSTEMS**

(Electrical &amp; Electronics Engineering)

Time: 3 hours

Max. Marks: 70

**PART – A**  
(Compulsory Question)

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- 1 Answer the following: (10 X 02 = 20 Marks)
- (a) Illustrate the difference between DSB-SC and DSB-FC. 2M
  - (b) Write the main functions of a radio transmitter. 2M
  - (c) Draw the waveform of FM. 2M
  - (d) What is the advantage of FM demodulation over AM demodulation in terms of noise performance? 2M
  - (e) Define a band-pass signal and give an example. 2M
  - (f) How does oversampling affect the quality of a reconstructed signal? 2M
  - (g) What is the principle behind Binary Phase Shift Keying (BPSK)? 2M
  - (h) Discuss how does a regenerative repeater differs from an analog repeater. 2M
  - (i) List the role of the base station in mobile communication. 2M
  - (j) Mention the concept of IP addressing in computer networks. 2M

**PART – B**  
(Answer all the questions: 05 X 10 = 50 Marks)

- 2 Demonstrate Double Sideband Full Carrier (DSB-FC) modulation, and derive its mathematical expression. 10M
- OR**
- 3 Describe the concept of frequency translation and why it is necessary in communication systems. Discuss the process of up-conversion and down-conversion with examples. 10M
- 4 Explain the concept of Angle Modulation. Compare Frequency Modulation (FM) and Phase Modulation (PM) in terms of their principles, waveforms, and applications. 10M
- OR**
- 5 Describe two different methods used for FM modulation. Also, explain the basic principle of FM demodulation and compare two demodulation techniques. 10M
- 6 Exemplify the concept of Pulse Amplitude Modulation (PAM) in detail. 10M
- OR**
- 7 Paraphrase the Sampling Theorem and its significance in signal processing. 10M
- 8 Describe the working of Binary Phase Shift Keying (BPSK) modulation. 10M
- OR**
- 9 What is a Regenerative Repeater and why is it important in digital communication systems? Elaborate. 10M
- 10 With the help of a block diagram, describe the working of an optical fiber communication system. Discuss the advantages of optical communication over traditional copper-based communication systems. 10M
- OR**
- 11 Construct the block diagram of a cellular mobile communication system. Discuss the working of different blocks and how the handoff process occurs during movement between cells. 10M

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